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1. CLO4 (20 คะแนน)

You are standing in the store trying to decide if you can afford a box of chocolate bars. When we buy an item in this town, there will be a sales tax of 8% added. You are lucky. During this period, every chocolate item in the store is on sale at 15% off.

For example, assume that a chocolate item in this store is $20. This means that the chocolate item is really $17. Then, after adding sales tax, the total cost after applying the sales tax will be $18.36.

Let f: R+ → R+ be a function that computes the price of an item after discount 15%.

Let g: R+ → R+ be a function that yields the price of an item if sales tax applied.

Define a function h based on the functions f and g such that the function h provides the total cost of a chocolate item you must pay.

let x is an original price of chocolate.

$f: R+ \to R+$ is a function that computes price of chocolate by discount the original price of chocolate by 15%.

so $f(x) = X(100-15)\% = X \cdot \frac{85}{100}$

$g: R+ \to R+$ is a function computes the price of product plus tax by 8%

so $g(y) = y(100+8)\% = y \cdot \frac{108}{100}$ | y is a price of product.

Therefore total price of one chocolate bar can compute by func. h by

$h: R+ \to R$ and $h(x) = (g \circ f)(x) = g(f(x))$

$g(f(x)) = g(x \cdot \frac{85}{100})$

$= x \cdot \frac{85}{100} \cdot \frac{108}{100}$

$= 0.918\, x$

summary : chocolate's price equal to 91.8% of their original price from func. h.

$h(x) = 0.918\, X$

2. CLO4 (10 คะแนน) Define a function f: $\mathbf{Z} \to \mathbf{Z} \times \mathbf{Z}$ by $f(x) = (2x+3,\ x-4)$.

Does f map $\mathbf{Z}$ onto $\mathbf{Z} \times \mathbf{Z}$? Prove or disprove.

To be onto function, they must be $(a,b) \in Z \times Z$ $\forall x \in Z \mid (2x+3, x-4) = (a,b)$

let $b = x-4$ then $x = b+4$ — ①

put ① on $2x+3$ | $2x+3 = 2(b+4)+3 = 2b+8+3$

$= 2b+11$

the equation will be true when $a = 2b+11$

Disprove $a = 2b+11$ let $(0,0) = (a,b)$

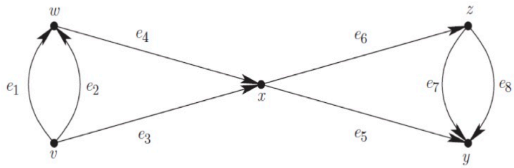
so $0 = 2(0)+11$

$0 = 11$

the equation be false so

func f. not a onto function.

3. CLO4 (15 คะแนน) Let G be the following directed graph.



Let V be the set of vertices in G and let E be the set of edges in G.

This graph G describes a function as follows.

$$i : E \longrightarrow V \times V$$

where $i(e) = (a, b)$ iff edge e goes from vertex a to vertex b.

(a) (2 คะแนน) Show the domain of function i (list all members of the set)

(b) (3 คะแนน) Show the co-domain of function i (list all members of the set)

(c) (4 คะแนน) List the values of $i(e_1)$, $i(e_2)$, …, $i(e_8)$.

(d) (3 คะแนน) Is the function i one-to-one? Why or why not?

(e) (3 คะแนน) Is the function i onto? Why or why not?

(a) $E = \{e_1, e_2, e_3, e_4, e_5, e_6, e_7, e_8\}$

(b) $\{(v,v),(v,w),(v,x),(v,y),(v,z),(w,v),(w,w),\dots,(z,z)\}$

(c) $i(e_1) = (v, w)$ $i(e_2) = (v, w)$
$i(e_3) = (v, x)$ $i(e_4) = (w, x)$
$i(e_5) = (x, y)$ $i(e_6) = (x, z)$
$i(e_7) = (z, y)$ $i(e_8) = (z, y)$

(d) no, because there has $\exists e_i$ such as e_1 and e_2, e_7 and e_8 is same begining point and end point.

(e) no, because members in Domain & Range of i can't be exceed 8 while the member of CO-Domain has $5 \times 5 = 25$ then there is no way to be onto function.

4. CLO4 (25 คะแนน) Define a relation on **Z** by $a\ R\ b$ if $a^2 = b^2$.

(a) (15 คะแนน) Prove that R is an equivalence relation.

(b) (10 คะแนน) Describe the equivalence classes.

(a) To prove Equivalence Relation must prove 3 propety {Reflexive, Symmetric, Transitive}

1. Reflexive : $\forall a \in Z \mid aRa$ Demand $a^2 = a^2$ is always true
   so aRa
2. Symmetric: $\forall a, b \in Z \mid aRb$ Demand $a^2 = b^2$ is always true
   so aRb
3. Transitive: $\forall a, b, c \in Z \mid aRb$ and bRc then prove aRc
   Demand $a^2 = b^2$, $b^2 = c^2$ then
   $a^2 = c^2$ so aRc

summary R has Equivalence Relation.

(b) $[a] = \{x \in Z \mid xRa\}$ | a is [a] is set of $x \in Z \mid xRa$

$[a] = \{x \in Z \mid x^2 = a^2\}$

$x^2 = a^2$ that $x = -a, x = a$

$[a] = \{a, -a\}$ #

5. CLO4 (10 คะแนน) Draw all equivalence classes based on the relation

"Congruence modulo 5" ($a \equiv b\ [\mathrm{mod}(5)]$)

Note. For each equivalence class, you may use the Ellipsis symbol (…). However, to avoid any ambiguity, you must specify enough number of members prior to using the Ellipsis symbol.

possible member = 0, 1, 2, 3, 4

$[0] = \{\ldots, -10, -5, 0, 5, 10, \ldots\}$

$[1] = \{\ldots, -9, -4, 1, 6, 11, \ldots\}$

$[2] = \{\ldots, -8, -3, 2, 7, 12, \ldots\}$

$[3] = \{\ldots, -7, -2, 3, 8, 13, \ldots\}$

$[4] = \{\ldots, -6, -3, 4, 9, 14, \ldots\}$

6. CLO4 (10 คะแนน) Let F be a relation on R (set of real numbers)

$$\forall a, b \in R,\ a\ F\ b \text{ if and only if } |a-b| < 1$$

Disprove that F is an equivalence relation.

$\forall a,b \in R \mid aFb \leftrightarrow |a-b| < 1$

Dis prove : Transitive

for every $a,b,c \in R$ if $aFb \wedge bFc \rightarrow aFc$ that $|a-b| < 1$

and $|b-c| < 1$ then $|a-c| < 1$

Counterexample: $|a-c| \geq 1$ prove this true

let $a=0$, $b=0.5$ and $c=1.2$

aFc : $|a-c| = |0-1.2| = 1.2$ and $1.2 \geq 1$

so F isn't an equivalence relation.

7. CLO4 (10 คะแนน) Define a relation on **Z** by setting $x\ R\ y$ if $x\ y$ is even.

(a) (5 คะแนน) Give a counterexample to show that R is not reflexive.

(b) (5 คะแนน) Give a counterexample to show that R is not transitive.

(a) give $x = 1$; $x \cdot x = x^2 = (1)^2 = 1$

by Definition for even number that

$x = 2n \mid n \in Z$

but there is no way to make $2n = 1$

so 1 isn't even number then $1 \not R 1$

R isn't reflexive.

(b). let $x = 1$, $y = 2$, $z = 3$

xRy : $xy = 1 \times 2 = 2$, 2 is even xRy is true

yRz : $yz = 2 \times 3 = 6$, 6 is even yRz is true

xRz : $xz = 1 \times 3 = 3$, 3 isn't even xRz is false.

R isn't transitive